Formalizing the Yu–Wang Quartic SPR Synthesis Theorem with a Single Perturbation Parameter

Zhengtai Xiong[1], Tiancheng Xu[1], Xiang Ji[1], Guowei Dou[1], Long Wang[2,3], Wensheng Yu[1]

[1]School of Electronic Engineering, Beijing University of Posts and Telecommunications, Beijing 100876, China

[2]Center for Systems and Control, College of Engineering, Peking University, Beijing 100871, China

[3]Center for Multi-Agent Research, Institute for Artificial Intelligence, Peking University, Beijing 100871, China

Abstract

We formalize the Yu–Wang SPR synthesis theorem for fourth-order polynomial segments in Lean. The established criterion states that two monic quartic denominators admit a common equal-degree strictly positive real numerator if and only if their whole convex segment is Hurwitz stable. The proof retains the original elliptic-region strategy. Explicit support functions establish intersection of the two regions, and a completed-square margin permits a single positive parameter to perturb the constant and leading coefficients simultaneously. The numerator can be chosen as $\varepsilon s^4+s^3+xs^2+ys+\varepsilon$, with an explicit admissible bound on ε. The converse is proved independently by an interlacing argument for numerator stability and by reciprocation. The Lean development imports only Mathlib and proves the theorem for the explicit quartic Routh–Hurwitz coefficient predicate. The contribution is a streamlined formalization with a quantitative perturbation step, not a new segment criterion.

Keywords: quartic polynomial segment; strictly positive real synthesis; formal verification; ellipse; perturbation.

1 Introduction and attribution

Following the cubic synthesis result of Yu and Huang [1], Yu and Wang proved that robust Hurwitz stability of a quartic polynomial segment is equivalent to the existence of a common equal-degree SPR numerator [2]. Their proof constructs an open ellipse for each denominator, proves intersection through a separating tangent line, obtains a cubic numerator by a small constant perturbation, and then raises its degree. This geometric route is the mathematical foundation of the present formalization.

The file `SPR/simplified/n4.lean` retains the ellipse and separation mechanism. It replaces tangent-line elimination by a dual quadratic identity, and it chooses the constant perturbation and leading coefficient to be the same positive number. These changes provide explicit inequalities convenient for a proof assistant. They do not change the original synthesis criterion or replace its principal geometric idea. Accordingly, this article is positioned as a formalization and refinement of [2].

The supplied arbitrary-order manuscript [3] guides the presentation, but its finite-contact proof is not invoked. The quartic Lean file is independent of the cubic, quintic, sextic, and arbitrary-order synthesis files. In particular, the existence theorem is proved within the quartic coefficient model rather than imported as a specialization of a more general result.

2 The coefficient model and synthesis statement

Let

$$a(s) = s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4, \qquad b(s) = s^4 + b_1 s^3 + b_2 s^2 + b_3 s + b_4, \qquad p_\lambda = (1-\lambda)a + \lambda b.$$

Define $H_4(a)$ by

$$a_j > 0 \ (1 \le j \le 4), \qquad a_1 a_2 - a_3 > 0, \qquad \Delta_a := a_1 a_2 a_3 - a_3^2 - a_1^2 a_4 > 0. \tag{2.1}$$

This is the classical quartic Routh–Hurwitz coefficient condition. In the formal file it is the definition of `Hurwitz`. The usual interpretation in terms of roots in the open left half-plane is external to this coefficient-model formalization; the file does not prove a general Routh–Hurwitz theorem. Notice that positivity of all coefficients and $\Delta_a > 0$ already imply $a_1 a_2 - a_3 > 0$.

For a real quartic c with positive leading coefficient, call c/a SPR when $H_4(a)$ holds and

$$\operatorname{Re} \frac{c(i\omega)}{a(i\omega)} > 0 \quad (\omega \in \mathbb{R}). \tag{2.2}$$

Put

$$E_a(t) = t^2 - a_2 t + a_4, \qquad O_a(t) = a_3 - a_1 t.$$

Then $a(i\omega) = E_a(\omega^2) + i\omega O_a(\omega^2)$. Since

$$a_1^2 E_a(a_3/a_1) = -\Delta_a < 0, \tag{2.3}$$

the even and odd parts cannot vanish together on the imaginary axis; $a_4 > 0$ handles $\omega = 0$. Thus the quotient in (2.2) is defined at every real frequency. Equal-degree monic quotients tend to 1 at infinity.

Theorem 2.1 (Formalized quartic segment criterion)**.** *Assume $H_4(a)$ and $H_4(b)$. The following are equivalent:*

1. *$H_4(p_\lambda)$ for every $\lambda \in [0,1]$;*
2. *a common monic quartic SPR numerator exists;*
3. *a common quartic SPR numerator with positive leading coefficient exists.*

If these conditions hold, there are $x, y \in \mathbb{R}$ and $\varepsilon > 0$ such that

$$c_\varepsilon(s) = \varepsilon s^4 + s^3 + x s^2 + y s + \varepsilon \tag{2.4}$$

works at both endpoints. The cubic $q_\varepsilon = s^3 + x s^2 + y s + \varepsilon$ also has strictly positive real part after division by either endpoint at every real frequency.

The reduced-degree statement is stated separately because the equal-degree SPR convention used here does not call q_ε/a a quartic SPR quotient.

3 The ellipse and its support function

For $x, y \in \mathbb{R}$ define

$$F_a(t) = (a_1 - x)t^2 + (a_2 x - a_1 y - a_3)t + (a_3 y - a_4 x), \tag{3.1}$$

$$C_a(x,y) = (a_2 x - a_1 y - a_3)^2 - 4(a_1 - x)(a_3 y - a_4 x), \tag{3.2}$$

$$\Omega_a = \{(x,y) : C_a(x,y) < 0\}. \tag{3.3}$$

Expansion of (3.2) yields

$$C_a(x,y) = (a_2^2 - 4a_4)x^2 + 2(2a_3 - a_1a_2)xy + a_1^2y^2 \\ - 2(a_2a_3 - 2a_1a_4)x - 2a_1a_3y + a_3^2,$$

the ellipse polynomial of [2, Lemma 1]. Its center is $(a_1/2, a_2/2)$. With $X = x - a_1/2$, $Y = y - a_2/2$,

$$C_a(x,y) = H_a(X,Y) - \Delta_a,$$

where

$$H_a(X,Y) = (a_2^2 - 4a_4)X^2 + 2(2a_3 - a_1a_2)XY + a_1^2Y^2.$$

The identity

$$a_1^2 H_a(X,Y) = \big(a_1^2 Y + (2a_3 - a_1a_2)X\big)^2 + 4\Delta_a X^2 \tag{3.4}$$

shows that H_a is positive definite and Ω_a is a nonempty open convex ellipse. It also gives $|X| < a_1/2$ on Ω_a, so $a_1 - x > 0$. Completing the square in (3.1) gives

$$F_a(t) \geq m_a := \frac{-C_a(x,y)}{4(a_1 - x)} > 0 \qquad (t \in \mathbb{R}). \tag{3.5}$$

For a nonzero normal (A,B) introduce

$$D_a(A,B) = a_1^2A^2 - 2(2a_3 - a_1a_2)AB + (a_2^2 - 4a_4)B^2, \qquad r_a = \sqrt{D_a(A,B)} > 0.$$

The determinant of the matrix of H_a is $4\Delta_a$. Consequently the support values of the open ellipse are

$$\sup_{\Omega_a}(Ax + By) = \frac{a_1A + a_2B + r_a}{2}, \qquad \inf_{\Omega_a}(Ax + By) = \frac{a_1A + a_2B - r_a}{2}. \tag{3.6}$$

The Lean proof constructs boundary support points and approaches them from the center, so it does not assume that a supremum is attained inside an open ellipse.

Lemma 3.1. *If the entire quartic segment satisfies H_4, then $\Omega_a \cap \Omega_b \neq \varnothing$.*

Proof. Otherwise separation and (3.6) provide a nonzero normal with

$$r_a + r_b \leq L := (b_1 - a_1)A + (b_2 - a_2)B.$$

Direct expansion, including the term $-4a_4B^2$, gives

$$D_{p_\lambda}(A,B) = (1-\lambda)D_a(A,B) + \lambda D_b(A,B) - \lambda(1-\lambda)L^2. \tag{3.7}$$

The fourth coefficient contributes affinely, so no additional quadratic remainder occurs. At $\lambda_* = r_a/(r_a + r_b)$, the weighted sum of the first two terms equals r_ar_b, while $\lambda_*(1-\lambda_*)L^2 \geq r_ar_b$. Thus $D_{p_{\lambda_*}}(A,B) \leq 0$, contradicting the positive definiteness supplied by $H_4(p_{\lambda_*})$. □

This replaces the common-tangent coordinate calculation of [2, Lemma 2]. Both arguments pass from disjoint elliptic regions to failure of stability at a convexly interpolated denominator; the proof strategy remains the same.

4 A single perturbation parameter

Choose $(x, y) \in \Omega_a \cap \Omega_b$. For a general numerator

$$c(s) = rs^4 + s^3 + xs^2 + ys + z$$

the cross polynomial is

$$Q_{a,c}(t) = tF_a(t) + zE_a(t) + rt^2E_a(t), \qquad \operatorname{Re}\frac{c(i\omega)}{a(i\omega)} = \frac{Q_{a,c}(\omega^2)}{|a(i\omega)|^2}. \tag{4.1}$$

The following estimate removes the need for two successive smallness choices.

Lemma 4.1. *For $t \geq 0$,*

$$E_a(t) \geq a_4 - a_2t, \qquad t^2E_a(t) \geq -a_2^3t. \tag{4.2}$$

If $r, z \geq 0$ and $za_2 + ra_2^3 \leq m_a$, then $Q_{a,c}(t) \geq za_4$ for all $t \geq 0$.

Proof. The first inequality follows from $t^2 \geq 0$. For the second, if $t \geq a_2$ then $E_a(t) = t(t - a_2) + a_4 > 0$. If $0 \leq t \leq a_2$, then

$$tE_a(t) = t^3 - a_2t^2 + a_4t \geq -a_2t^2 \geq -a_2^3.$$

Multiplication by $t \geq 0$ proves the second bound. Combining these estimates with (3.5) in (4.1) yields

$$Q_{a,c}(t) \geq za_4 + (m_a - za_2 - ra_2^3)t \geq za_4.$$

□

Set

$$\varepsilon = \min\left\{\frac{m_a}{a_2 + a_2^3}, \frac{m_b}{b_2 + b_2^3}\right\} > 0. \tag{4.3}$$

Both denominators in (4.3) are positive. Taking $r = z = \varepsilon$ gives

$$Q_{a,c_\varepsilon}(t) \geq \varepsilon a_4 > 0, \qquad Q_{b,c_\varepsilon}(t) \geq \varepsilon b_4 > 0 \quad (t \geq 0).$$

Thus (2.4) is a common quartic SPR numerator. Equality in the smallness bound is allowed: the strictly positive constant term εa_4 remains. The same ε, with $r = 0$, gives the common cubic q_ε. Finally, $c_\varepsilon/\varepsilon$ is the required monic quartic.

Remark 4.2. Formula (4.3) is an admissible bound after choosing a common ellipse point. It is not claimed to maximize a passivity margin or to be invariant under a change of frequency units. The proof establishes feasibility for the fixed polynomial coordinates of the theorem.

5 Necessity by a numerator-stability argument

Lemma 5.1. *If $H_4(a)$ and a monic quartic n has $\operatorname{Re}(n(i\omega)/a(i\omega)) > 0$ at every real frequency, then $H_4(n)$.*

Proof. By (2.3), the monic quadratic E_a has two positive roots α, β satisfying

$$0 < \alpha < a_3/a_1 < \beta.$$

At these roots the cross polynomial reduces to $tO_n(t)O_a(t)$. Strict positivity implies

$$n_3 - n_1\alpha > 0, \qquad n_3 - n_1\beta < 0.$$

Therefore $n_1 > 0$, $n_3 > 0$, and $u = n_3/n_1 \in (\alpha, \beta)$. At zero frequency, positivity gives $n_4 > 0$. Since $E_a(u) < 0$ and $O_n(u) = 0$, cross positivity at u forces

$$E_n(u) = u^2 - n_2 u + n_4 < 0.$$

This implies $n_2 > 0$ and, after multiplication by n_1^2,

$$n_1 n_2 n_3 - n_3^2 - n_1^2 n_4 > 0.$$

Together with coefficient positivity these are exactly (2.1) for n. □

If a common monic numerator n exists, Lemma 5.1 makes n a stable denominator. Reciprocation preserves strict positivity of the real part, since

$$\operatorname{Re}\frac{1}{z} = \frac{\operatorname{Re} z}{|z|^2} \quad (z \neq 0).$$

Consequently a/n and b/n have positive real parts, and so does p_λ/n by linearity. Applying Lemma 5.1 again proves $H_4(p_\lambda)$ for every λ. Arbitrary positive leading coefficients reduce to the monic case by scaling. This completes Theorem 2.1.

Corollary 5.2. *The same numerator works for every denominator p_λ on the segment.*

Proof. The cross product $\operatorname{Re}(c(i\omega)\overline{p_\lambda(i\omega)})$ is the convex combination of the two strictly positive endpoint cross products. Divide by $|p_\lambda(i\omega)|^2 > 0$. □

6 Formal verification

The proof presented in this paper has been formally verified in Lean. The code repository is `https://github.com/XPiPiT/n_random/tree/main/SPR/simplified`.

7 Conclusion

The quartic synthesis theorem admits a compact formalization that preserves the original ellipse geometry. A dual support identity proves intersection, and the inequality $t^2 E_a(t) \geq -a_2^3 t$ gives a simultaneous perturbation of the constant and leading coefficients. The result supplies both reduced-degree and equal-degree witnesses, together with an independent formal necessity proof.

References

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